

3.000 - Instron Station

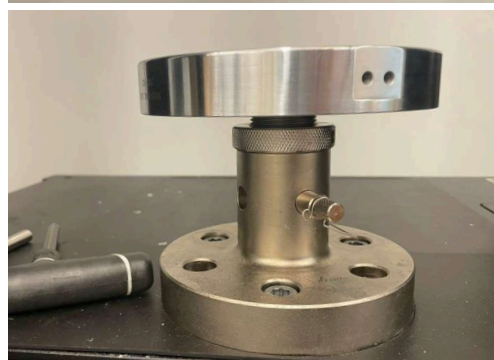
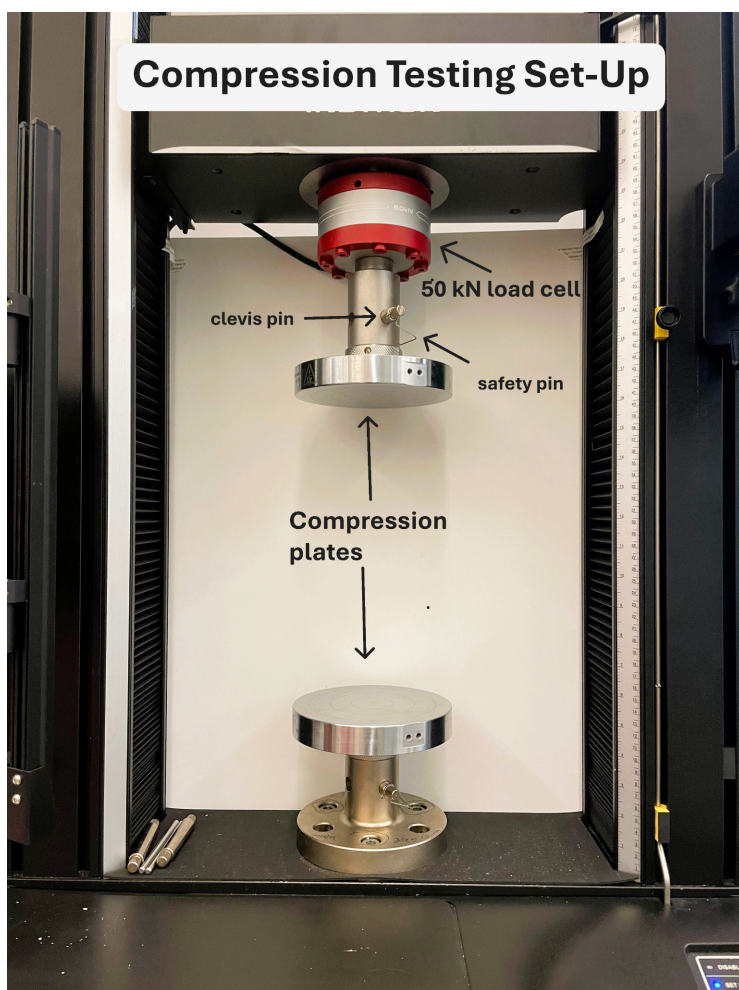
In this rotation you will crush a roasted and a green bean using the Instron universal testing system. You will record the highest force sustained by each bean before cracking. *You will need to save two values: the highest force sustained by the green bean and by the roasted bean. You will submit these to Canvas once this rotation is complete.*

Instructions:

1. Log on to the instrument workstation using your MIT Kerberos.



2. Open the "Bluehill Universal" software.
3. If the event log opens up, you can click close.
4. Put together a compressive testing set up. You will need two compression plates, two clevis pins, and two safety pins. Attach the pieces according to the photos below.
 - a. To attach the pieces, align the internal holes and insert a clevis pin. Then add a safety clip and rotate the metal ring to secure it.



5. Place a coffee bean at the center of the bottom compression plate.
6. In Bluehill Universal, click Test. From here, you can run a basic “QuickTest” or choose from any method.
7. Click 3.000 Coffee Bean.
8. When prompted, set travel limits to prevent the head and attached plates from colliding into anything. Click OK when done.
9. Using the hand controller, jog the head until it’s about a cm above your coffee bean.

Controller when Instron is in set-up mode

These buttons duplicate functions assigned to Softkey 1 and 2 in the software



10. **STOP HERE.** Call over an instructor to check your setup and clear you to start.

Before you start, read this: The method is set up to stop once it detects a sudden 30% decrease in force, i.e. when the bean breaks. In case this does not happen, you need to be ready to stop the test. To stop the test, you can push down the big red emergency button

on the machine or the stop button on the hand controller (). Before you start, make sure you know where these buttons are. Do NOT let the compression plates hit each other.



11. Zero the displacement by clicking () on the hand controller.
12. Using the hand controller, click unlock () and start () in quick succession.
13. Once the bean cracks, stop the test.
14. Write down the highest force withstood by the bean before cracking.
15. Replace the bean with the one you haven't tested, and repeat the test.
16. Once done, disassemble your testing set up and put the compression plates back into the boxes they came from.
17. Close Bluehill Universal and sign out of your account.